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Optimum Allocation of Weight Between Two Stages (rev. 1)

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Introduction

A frequently asked question is, What's the optimum allocation weight between two stages? The answer can be easily found once we decide the optimum criterion is minimum cost, or by a common assumption, minimum total weight. (Yes, it's still true, we buy 'em by the pound, sailboats, cars, rockets, etc.)

In a general sense there are many techniques to solve this problem. One can always directly evaluate the objective function (ideal velocity gained) for various combined weights. This provides insight into the nature of the optimum, but no guaranteed values for the stationary points. Classical calculus techniques (differentiate the objective function with respect to stage weights, and set the derivatives to zero) will work, but leads to very messy algebra, especially for more than two stages. But, the best is the lagrangian multiplier approach of refs. (1) and (2). This has been recounted subsequently, e. g., ref's. (3) and (4), but because these old documents are difficult to find, it is recapitulated here. The great beauty of this approach is that it is very easy to use, even with many stages.

Nomenclature

Mnemonic	Definition
i	Stage index. Stage 1 is at the bottom of the stack.
c_i	Exhaust velocity of burnt propellant relative to the ith stage $= I_{SPi} g_0$,
M_{PL}	Payload mass,
M_{Ui}	Throw weight of the ith stage. $M_{UN} = M_{PL}$, and $M_{Ui} = M_{0i+1}$
M_{PRi}	Useful (not including trapped ullage) propellant mass of the ith stage,
M_{STi}	Structural mass of the ith stage = Inert mass of the ith stage,
M_{0_i}	Loaded mass of the ith stage + loaded masses of all higher stages + payload mass,
I_{SPi}	Specific impulse of the ith stage,
g_0	Reference acceleration of gravity,
N	Number of stages in a rocket assembly,
ΔV_i	Ideal velocity increase during burning of the ith stage,
ΔV_{IDi}	Ideal burnout velocity of the ith stage
ΔV_{TOT}	Ideal velocity increase due to operation of all stages,
λ	Lagrangian multiplier,

Mnemonic	Definition
φ_i	Propellant ratio = $\frac{M_{PRi}}{M_{Oi}}$,
η_i	Payload ratio = $\frac{M_{Oi+1}}{M_{Oi}}$,
μ_i	Mass ratio = initial mass / burnout mass of the ith stage, $=\frac{M_{0i}}{M_{0i} - M_{PRi}}$,
σ_i	Structural efficiency of the ith stage = $M_{STi} / M_{STi} + M_{PRi}$

Glossary

- See Fig. 1 for a notional description of a three stage rocket.
- Ideal velocity is the velocity increase that would have been obtained if there were no drag or gravity losses,
- Initial mass of the ith stage is the mass, with propellant, of the ith stage + the total mass of all stages above the ith stage, with propellant, + payload mass,
- Burnout mass of the ith stage is the structural mass of the ith stage + the total mass of all stages above the ith stage, with propellant, + payload mass
- Reference acceleration of gravity is that official value of g used to convert measured weights to masses,
- Structural mass of a stage is all non-consumable mass = inert mass of the stage,
- Useful propellant mass of a stage is all the mass consumed and expelled from the stage

Tsiolkovskii's Law

The first modern model for the ideal velocity was due to K. E. Tsiolkovskii, a high school physics teacher living in a remote farming district in Czarist Russia. His result for a single stage rocket was

$$\Delta V = c \log \mu = I_{SP} g_0 \log(\frac{M_{ST} + M_{PR}}{M_{ST}}) .$$

(1)

Tsiolkovskii also showed that for a multistage rocket the total ideal velocity is just the sum of the individual stage ideal velocities:

$$\Delta V_{TOT} = \sum_1^N \Delta V_i = \sum_1^N I_{SPi} g_0 \log(\mu_i) .$$

(2)

Finally, if all stages have the same mass ratio and specific impulse,

$$\Delta V_{TOT} = N I_{SP} g_0 \log \mu$$

By using multiple stages very high total velocities can be attained.

Optimum Stage Masses

There are two ways to specify this problem:

- First, for a fixed total mass, number of stages and payload mass find that stage mass allocation that maximizes the ideal velocity increase.
- Second, for a fixed ideal velocity increase, number of stages and payload mass find that stage mass allocation that minimizes the total mass at liftoff.

Since these two formulations are essentially the same, we choose arbitrarily to maximize the total ideal velocity

Next, it is easy to show that

$$\varphi_i = (1 - \sigma_i)(1 - \eta_i) \text{ , and}$$

(3)

$$\mu_i = \frac{1}{1 - \varphi_i} = \frac{1}{\sigma_i (1 - \eta_i) + \eta_i},$$

(4)

The total payload ratio is

$$\frac{M_{UN}}{M_{01}} = \frac{M_{U1}}{M_{01}} \frac{M_{U2}}{M_{02}} \cdots \frac{M_{UN}}{M_{0N}} = \frac{M_{02}}{M_{01}} \frac{M_{03}}{M_{02}} \cdots \frac{M_{PL}}{M_{0N}} \text{ , or}$$

$$\eta_{TOT} = \frac{M_{PL}}{M_{01}},$$

(5)

and it's easy to show that

$$\frac{M_{PL}}{M_{01}} = \eta_1 \eta_2 \cdots \eta_N .$$

(6)

Next, the total propellant ratio is:

$$\varphi_{TOT} \equiv \frac{\sum M_{PRi}}{M_{01}} \text{ , and}$$

$$\frac{M_{PRi}}{M_{01}} = \frac{M_{PRi}}{M_{0i}} \frac{M_{0i}}{M_{0i-1}} \cdots \frac{M_{02}}{M_{01}} = \varphi_i \eta_1 \eta_2 \cdots \eta_{i-1}$$

Then, $$\varphi_{TOT} = \varphi_1 + \varphi_2\eta_1 + ... + \varphi_N\eta_1\eta_2...\eta_{N-1} \ .$$
(7)

In the same way, the total structural efficiency is

$$\sigma_{TOT} = \frac{\sum M_{STi}}{\sum M_{STi} + \sum M_{PRi}} = \frac{1}{1 - \eta_{TOT}} \sum \frac{M_{STi}}{M_0} .$$

But, $$M_{STi} = \frac{\sigma_{STi}}{1 - \sigma_{STi}} M_{PRi}, \text{ and}$$

$$\frac{M_{STi}}{M_0} = \frac{\sigma_i}{1 - \sigma_i} \varphi_i\eta_1\eta_2....\eta_{i-1} .$$

(8)
Finally,

$$\sigma_{TOT} = \frac{1}{1 - \eta_{TOT}} \left[\sigma_1(1 - \eta_1) + \sigma_2(1 - \eta_2)\eta_1 + ... + \sigma_N(1 - \eta_N)\eta_1\eta_2....\eta_{N-1}\right].$$

(9)

Our optimization problem can be formulated in three equivalent ways: (a) Maximize the payload mass for given ΔV_{TOT} and total liftoff mass, or (b) Minimize the liftoff mass for given ΔV_{TOT} and payload mass, or (c) Maximize ΔV_{TOT} for a given payload ratio. So, this implies that we need to determine the payload ratios such that

$$\Delta V_{IDi} = - g_0 \sum_{i=1}^{N} I_{SPi} \log(\eta_i + \sigma_i(1 - \eta_i))$$

(10)

subject to $$\Pi_{i=1}^{N} \eta_i = \eta_{TOT} .$$
(11)

It will prove convenient to rewrite the side condition, eq. (11), as

$$\sum_{i=1}^{N} Log(\eta_i) - \log(\eta_{TOT}) = 0$$

(12)

Next, define an augmented function F as

$$F(\eta_i) = \Delta V_{IDi} + \lambda\left[\sum_{i=1}^{N} \log(\eta_i) - \log(\eta_{TOT})\right],$$

(13)

where λ (ref. (5)) is a lagrangian multiplier. The necessary condition for ΔV_{IDi} to be maximized while satisfying eq. (12) is

$$\frac{\partial F}{\partial \eta_i} = 0, i = 1,2,...,N .$$

(14)

Substituting eq's. (10) and (13) into eq. (14) gives

$$g_0 I_{SPi} \frac{1 - \sigma_i}{\eta_i + \sigma_i(1 - \eta_i)} - \frac{\lambda}{\eta_i} = 0.$$

(15)

Solving for the η_i leads to

$$\eta_i = \frac{\lambda\sigma_i}{(g_0 I_{SPi} - \lambda)(1 - \sigma_i)}.$$

(16)

The lagrangian multiplier can be found by substituting eq. (16) into eq. (12):

$$\sum_{i=1}^{N} \log\left[\frac{\lambda\sigma_i}{(g_0 I_{SPi} - \lambda)(1 - \sigma_i)}\right] = \log(\eta_{TOT}).$$

(17)

Now, the important fact is that eq.(17) has only one unknown, λ. The optimal payload ratio, and the maximum ΔV_{TOT} are found by solving eq. (17) for λ, and substituting the result into eq's. (16) to find the payload ratios. Finally, the ideal velocity can be found from eq.(10).

Note that eq.(17) has, in general, multiple solutions. It is necessary to find all the solutions and determine the one with the greatest ΔV_{TOT} by substitution.

Common Specific Impulse

In general, eq. (17) is an algebraic equation of order N that cannot be solved analytically. Its solution is straightforward if all stages share a common specific impulse. Then,

$$\frac{\lambda}{g_0 I_{SP} - \lambda} = \eta_{TOT} \left[\frac{(1-\sigma_1)(1-\sigma_2)....(1-\sigma_N)}{\sigma_1 \sigma_2 .... \sigma_N} \right]^{1/N}.$$

Next, substitute this into eq. (16) to give

$$\eta_i = \frac{\sigma_i}{1-\sigma_i} \left[\eta_{TOT} \frac{(1-\sigma_1)(1-\sigma_2)....(1-\sigma_N)}{\sigma_1 \sigma_2 .... \sigma_N} \right]^{(1/N)}.$$

(18)

Substitution of these results back into eq. (10) yields the maximum ideal velocity:

$$\Delta V_{ID\max} = -g_0 I_{SP} \log \left[\sigma_1 \sigma_2 .... \sigma_N \left[1 + \left[\eta_{TOT} \frac{(1-\sigma_1(1-\sigma_2)....(1-\sigma_N)}{\sigma_1 \sigma_2 .... \sigma_N} \right]^{(1/N)} \right]^N \right].$$

(19)

On the other hand, if ΔV_{ID} is specified

$$\eta_{TOT\max} = \frac{\left[\exp(-U/N) - (\sigma_1 \sigma_2 .... \sigma_N)^{(1/N)} \right]^N}{(1-\sigma_1)(1-\sigma_2)....(1-\sigma_N)},$$

where

$$U = \frac{\Delta V_{ID}}{g_0 I_{SP}}$$

(20)

For a two stage rocket, these simplify even more to

$$\Delta V_{ID\max} = -2\log\left[\sqrt{\sigma_1 \sigma_2} + \sqrt{\eta_{TOT}(1-\sigma_1)(1-\sigma_2)} \right], \text{ and}$$

(21)

$$\eta_{TOT\max} = \frac{\left[\exp(-U/N) - \sqrt{\sigma_1 \sigma_2} \right]}{(1-\sigma_1)(1-\sigma_2)}.$$

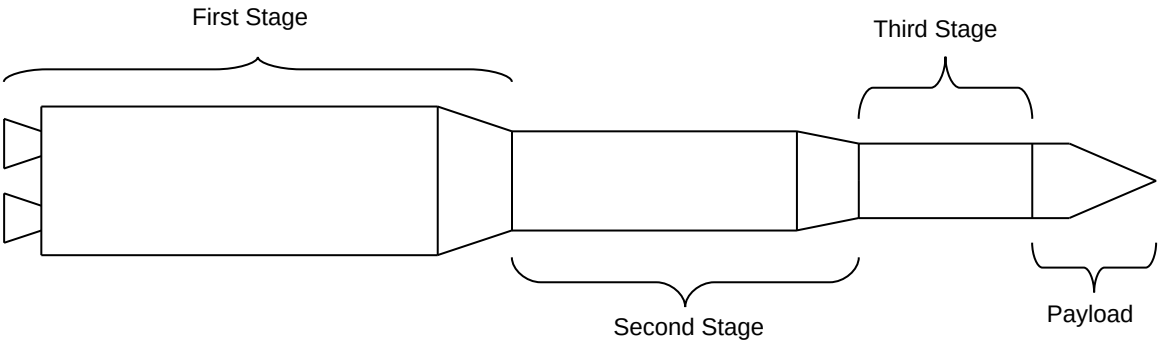
(22)

The individual stage payload ratios are found from eq. (18):

$$\eta_i = \frac{\sigma_i}{1-\sigma_i} \left[\eta_{TOT} \frac{(1-\sigma_1)(1-\sigma_2)}{\sigma_1 \sigma_2} \right]^{1/2}.$$

(21)

Assuming $I_{SP} = (I_{SP1} + I_{SP2})/2$, Eq. (22) has been coded as OPT2STGE.xls)



References

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